

10.7 Measures of Center and Spread

Lesson Introduction

 Benchmark	MA.6.DP.1.6 Given a real-world scenario, determine and describe how changes in data values impact measures of center and variation.		 Learning Target	I can determine which measures of center and variation best describe a distribution.
 Benchmark Coherence	Building On MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.		Working Toward MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.	
 Essential Questions	- For a symmetrical graph, what is the best measure of center? - For a symmetrical graph, what is the best measure of variation? - For a skewed graph, what is the best measure of center? - For a skewed graph, what is the best measure of variation?			
 Lesson Narrative	In this lesson, students will use what they know about mean, median, range, and IQR to determine the best measure of the center and variation to summarize the distribution. Students will be given real-world data and scenarios to use and apply their knowledge to choose the best measures of center and variation to describe data.			
 Component Summary	10.7.1 Warm-Up (~5 min.)	Notice/Wonder about data in different formats (<i>SE p. 231</i>)	10.7.4 Your Turn! (5-10 min.)	Determine the best measure of center and variation of given data sets in various forms (<i>SE p. 234</i>)
	10.7.2 Exploration (15-20 min.)	Discover how to find best fit measurement of variation and center (<i>SE p. 232</i>)	10.7.5 Wrap-Up (~5 min.)	Reflect on understanding of lesson (<i>SE p. 235</i>)
	10.7.3 Try It (5-10 min.)	Find the measure of the center of a data set (<i>SE p. 234</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Symmetrical - Skewed		(Optional) Paper	N/A

 Teacher Tips	Common Misconceptions - Students may confuse which would be a better measure of the center for a data set—mean or median. - Students may incorrectly think the variation can only be found in the range.	Things to Consider Encourage students to format the data in a way they can best read and find the measurements needed.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder The 10.7.1 Warm-Up positions students to engage in a Notice and Wonder routine. Through this routine, students analyze the similarities and differences between graphical representations of data and consider any questions they may have based on the visuals.	MA.K12.MTR.2.1: Multiple Representations Students will determine the measure of center and the measure of variation using data shown in multiple different ways. Different sets of data may need to be graphed or sorted to find answers.
 Differentiate Instruction	Notice 10.7.2 Exploration positions students to engage with the content through a variety of entry points. Question 1 displays the data visually. Parts a through c encourage visual cues to analyze the data, while part d encourages verbal processing through partner discussion. Reading through the blue boxes aloud can provide auditory entry points and help students synthesize the material.	
Lesson Notes:		

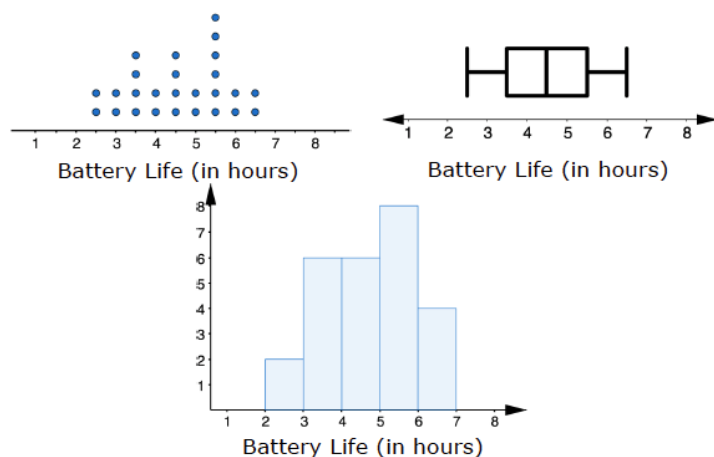
10.7.1 Warm-Up: Notice and Wonder

Component Overview: In this Warm-Up, students will discuss what they notice and wonder about the data shown in multiple formats.



10.7.1 Warm-Up: Notice and Wonder

The dot plot, histogram, and box plot summarize the hours of battery life for 26 cell phones that are constantly streaming videos.



Notice and Wonder

Engage students in a Notice and Wonder routine.

Encourage students to record similarities and differences they notice between the line plot, histogram, and box plot. Ask students what this makes them wonder about the data set or the data displays.

1. What do you notice?

Answers will vary. The dot plot and the histogram look the most similar. The median of the box plot looks to be very close to the middle.

2. What do you wonder?

Answers will vary. I wonder if all the batteries are the same. I wonder if the age of the phone affects the battery life.

10.7.2 Exploration: Measures of Center

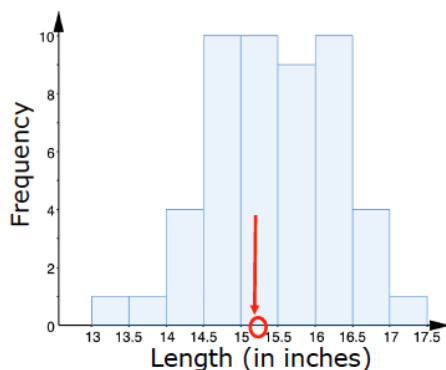
Component Overview: In this exploration, students work with a partner to determine the measures of center and spread that best describe the data presented in two different histograms. The first histogram is relatively symmetrical, while the second is skewed to the right. Students analyze the shape of the histograms to draw conclusions about how outliers impact the data sets. Students will also evaluate the best measures of center and spread to use when describing symmetrical versus skewed data sets.



10.7.2 Exploration: Measures of Center

1. Consider the histogram of the lengths of female sharks.

Shark Length



- a. Describe the shape of the histogram.
The graph is symmetrical, with most sharks having a length between 14.5 and 16.5 inches.
- b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of the data.
Shown on graph.
- c. Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.
Shown on graph.
- d. Compare your answers to B and C with your partner's answers. Discuss your reasoning, and if necessary, make changes.



The histogram provides a visual entry point. Consider having students further distinguish between median and mean by using different colored pencils to label each on the graph.



In a symmetrical graph, the mean and the median are located in the middle of the graph, approximately in the same location.

10.7.2 Exploration: Measures of Center (continued)

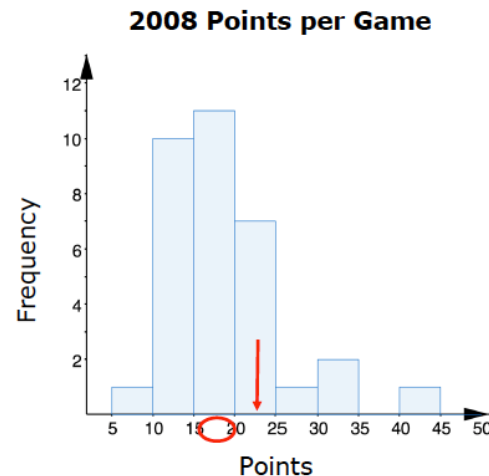
- e. The mean and the median of this histogram are located between 15 – 15.5. Discuss with your partner which measure most accurately describes the center of the histogram. Summarize your discussion.

Answers will vary. Both measures have about the same value, but the mean best describes the data because the average represents a typical length.



In a symmetrical graph with no outliers, the **mean** is the preferred measure of center and the **range** is the preferred measure of variation to describe the distribution.

2. The histogram displays the number of points per game scored by a college basketball player during the 2008 season.



- a. Describe the shape of the histogram.
Skewed to the right.



To solidify conceptual understanding, read the blue boxes as a class and ask the following questions:



- Are the values for mean and median usually close together?
- What could cause them to be farther apart?
- Why is the mean the preferred measure of center when a data set has no outliers?
- Why is the range the preferred measure of variation to use with mean?



Consider allowing students to draw pictures to match the words in the second blue box for those who would benefit from a visual of the statement.

10.7.2 Exploration: Measures of Center (continued)

- b. Circle the value on the horizontal axis of the graph that shows the approximate location of the median of this data.

Shown on graph.

- c. Draw an arrow ($\uparrow$) on the graph to indicate the approximate location of the mean of the data.

Shown on graph.

- d. In a symmetrical graph, the mean is the preferred measure of center to use. The histogram for points scored is skewed. Discuss with your partner why the mean would not be a good representation of the center for points scored. Summarize your discussion.

Answers may vary. The value of the mean is affected by the higher values that are causing the shape to be skewed. Using the mean may falsely represent the center as higher than it actually is.

- e. In this graph, the median is located between 15 – 20 and the mean is in the 20 – 25 interval. Which measure most accurately describes the center of the histogram?

The median because the peak and center of the data is closer to 15-20.

- f. Which measure of variation should be used to describe the distribution?

Answers will vary. The range is affected by the few large values that are causing the distribution to be skewed to the right, but they do not affect the IQR. The IQR should be used to describe the variation.



In a skewed graph or a graph with outliers, the **median** is the preferred measure of center and the **IQR** is the preferred measure of variation to describe the distribution.



For a digital learning extension, consider allowing students to enter each number into an Excel spreadsheet and create a chart with the data. Students can then find the mean and median of the given data using the equation functions in Excel.



Students may not understand why the mean would not be a good representation of the center in a skewed graph. To further scaffold this section, ask guiding questions if needed:

- *What do you know about outliers? How do they impact data distributions?*
- *Is the mean impacted by high/low outliers? How?*
- *Is the median impacted by outliers? Why or why not?*



Note that the values of the IQR and range cannot be determined since only the histogram is given. A box plot or data set would need to be included to determine those exact values.

10.7.3 Try It: Measures of Center

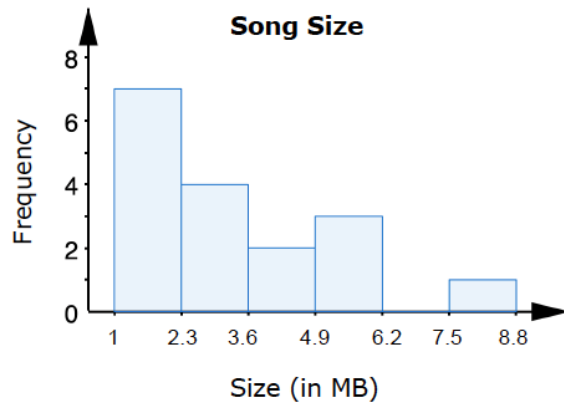
Component Overview: Students work independently to describe the shape of a histogram and determine which measure of center would best describe the data set and explain their choice.



10.7.3 Try It: Measures of Center

1. Nelson is interested in how much storage space his downloaded music takes up. The size of 17 randomly selected files is listed, in megabytes (MB), and displayed in a histogram.

1, 1.3, 1.3, 1.9, 1.9, 2.1, 2.2, 2.4, 2.5, 2.7, 3, 4.4, 4.7, 5, 5.6, 6, 7.5



- a. Describe the shape of the histogram.
Skewed right. The data peaks in the interval between 1 and 2.3.
- b. Which measure of center best describes the center of the histogram?
Median
- c. Explain your choice of center.
Answers will vary. Since the data skews right, using a median to measure the center makes sense since the mean is potentially affected by the outlier in the last interval.



Challenge students who finish early to explain which measure of variation would best describe the data set and why.

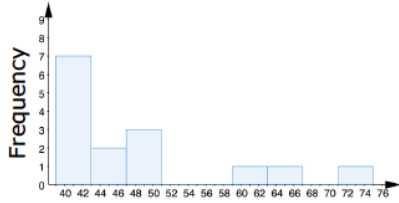
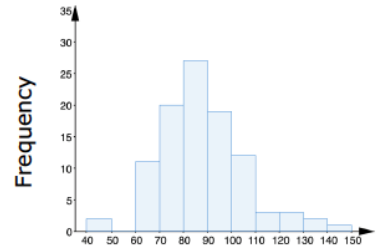
10.7.4 Your Turn!

Component Overview: In 10.7.4 Your Turn!, students determine the best measures of center and variation to describe data sets presented in multiple ways.



10.7.4 Your Turn!

- Determine which measures of center and variation best describe each data set.

Data	Measure of Center	Measure of Variation
Weekly biking times, in hours: 1,2,2,3,3,3,4,4,4,4,5,5,6,6,7	☐ Mean ☐ Median	☐ IQR ☐ Range
Daily high temperatures, in °F: 61, 61, 62, 64, 66, 67, 67, 67, 68, 69, 70, 70, 70, 71, 71, 72, 74, 74, 74, 75, 75, 75, 76, 76, 77, 78, 78, 79, 79, 95	☐ Mean ☐ Median	☐ IQR ☐ Range
2019-2020 NCAA Women's Division 1 Soccer Top Scorers 	☐ Mean ☐ Median	☐ IQR ☐ Range
Text Messages Sent 	☐ Mean ☐ Median	☐ IQR ☐ Range



Position students to predict what they think the answer should be for the first and second set and create a histogram to determine the spread of the data. Creating histograms to represent the data will provide additional visual entry points.

10.7.5 Wrap-Up: Reflection

Component Overview: In this Wrap-Up, students will reflect on their understanding of the learning target. Encourage students to take time to think about their understanding to target what they need to review and work on.



10.7.5 Wrap-Up: Reflection

1. Draw an **X** on the line to indicate where your current learning is related to each target.

I can determine which measures of center and variation best describe a distribution.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this,
and I feel confident.



Answers will vary.



Consider drawing the line and label across the entire whiteboard and asking students to stand in front of the label that describes their understanding of the lesson. This will help engage kinesthetic learners.